

1. Intelligent Agents in Smart Home Automation:

Introduction:

An intelligent agent is a system that observes its environment through sensors and performs action using actuators to achieve a goal.

1. Simple Reflex Agent:

Acts based on current inputs.

eg: switches on lights when motion is detected.

It is fast but cannot remember past events.

Advantage: Simple, fast, and easy to implement.

Disadvantage: Cannot remember past events or learn user preferences.

2. Model-Based Agent:

uses current inputs and past information.

eg: switches off lights when nobody is in the room.

Advantage: uses past information to make decision works better in partially observable environments.

Disadvantage: Requires memory and a more complex design.

3> Goal - Based Agent

A goal-based agent selects actions to achieve a specific goal.

eg: It adjusts the air conditioner to maintain the desired room temperature.

Advantage: supports planning and goal-oriented decisions.

Disadvantage: Requires more computation and planning time.

4> Utility - Based Agent:

A utility based agent selects the action that provides the greatest overall benefit by considering different factors.

eg: It maintains a comfortable temperature while reducing electricity consumption and maintaining security.

Advantage: Balances comfort, energy efficiency, and security.

Disadvantage: Designing a suitable utility function is difficult.

Comparison:

Agent	Main Function
Simple Reflex	uses current input.
Model - Based	uses current input & internal state.
Goal - Based	Achieves a specific goal
Utility - Based	Maximizes overall benefits.

Most effective Approach:

A hybrid approach combining all four agents is most effective. The simple reflex agent responds quickly to sensor inputs, the model-based agent remembers home conditions, the goal-based agent achieves desired objectives, and the utility-based agent balances comfort, energy efficiency, and security.

conclusion;

Intelligent agents make smart homes more comfortable, secure, and energy-efficient. A hybrid approach provides better results than using a single type of agent.

② Uniform cost Search (UCS) and Iterative Deepening Search (IDS) in a Maze.

Introduction:

Uninformed search algorithms explore possible paths without using additional information about the distance to the goal.

* Uniform cost search (UCS)

UCS expands the node with the lowest total path cost. It uses a priority queue to select the next node.

working:

- * Start from the initial position.

- * Select the lowest-cost path.

- * check for the exit.

- * Repeat until the exit is found.

Advantages: Finds the least-cost path.

Disadvantage: Requires more memory and time.

2) Iterative Deepening Search (IDS)

IDS repeatedly searches the maze by increasing the depth limit until the exit is found.

Advantage: Uses less memory and finds the shallowest solution when all step costs are equal.

Disadvantage: Repeats searches and may waste time.

Complexity:

UCS: Exponential time and space in the general case.

IDS: Time ($O(b^d)$), space ($O(bd)$)

Here (b) is the branching factor and (d) is the solution depth.

Intelligent Agents:

- * Simple Reflex: Acts on current inputs.
- * Model-Based: Remembers visited positions.
- * Goal-Based: plans to reach the exit.
- * Utility-Based: chooses paths based on cost and energy.

Best approach:

A model-based, goal-based agent is suitable.

VCS is best for different movement costs, while IDS is suitable for equal movement costs and limited memory.

Conclusion:

VCS finds the least-cost path, whereas IDS saves memory. The appropriate algorithm helps the robot navigate the maze efficiently.